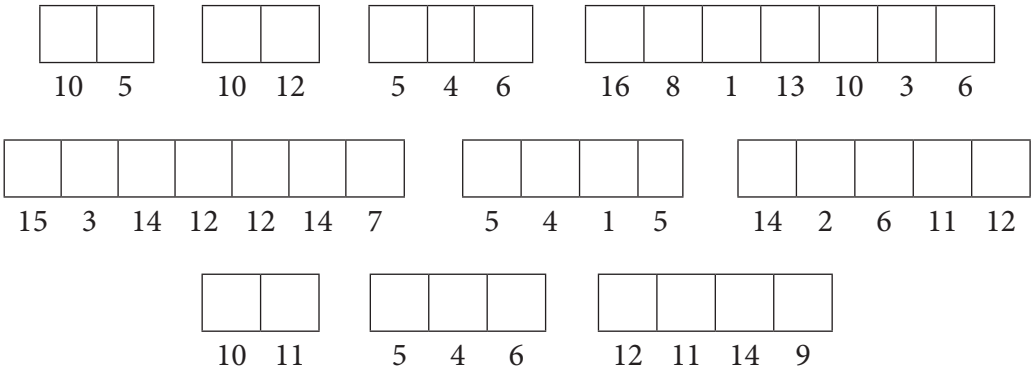
Name: _____

WORKSHEET

Geometric progressions 2

Courage is not the towering oak that sees storms come and go …



Alice M. Swaim

Rules

- In a geometric progression, the nth term t_n can be found by:

  $t_n = ar^{n-1}$

  where a is the first term and r is the common ratio.
- The common ratio can be found by:

  $r = \frac{t_n}{t_{n-1}}$.

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Questions

What is the common ratio in each of the following geometric progressions?

1 2, 10, 50, 250, …

2 $800,\ 200,\ 50,\ 12\frac{1}{2},\ \ldots$

3 81, 27, 9, 3, …

4 $\frac{1}{2},\ -1,\ 2,\ -4,\ \ldots$

5 What is the 15th term of the geometric progression 2, 6, 18, …?

6 What is the 9th term of the geometric progression $-\frac{1}{5},\ 1,\ -5, E\ldots$?

7 What is the 11th term of the geometric progression 1024, 512, 256, …?

8 What is the 7th term of the geometric progression $1\frac{1}{3},\ 1,\ \frac{3}{4},\ \ldots$?

How many terms are in each of the following geometric progressions?

9 $\frac{2}{3},\ 2,\ 6,\ \ldots\ 4374$

10 10 240, 5120, 2560, … 5

11 −7, 14, −28, … 3 670 016

12 20, 2, 0.2, … 0.000 000 000 02

13 Find the 8th term of a geometric progression if the 3rd term is 1 and the 18th term is 32 768.

14 Find the 6th term of a geometric progression if the 2nd term is −1 594 323 and the 11th term is 81.

15 Find the 11th term of a geometric progression if the 5th term is 4 and the 8th term is 256.

16 Find the 9th term of a geometric progression if the 7th term is 9375 and the 12th term is −29 296 875.

Solutions

A 5

B 16 384

E −78 125

F 234 375

G 32

H −2

I 12

L $\frac{1}{3}$

M 1

N 20

O −19 683

P $\frac{1}{4}$

Q $\frac{243}{1024}$

S 13

T 9 565 938

W 9

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